INTERNATIONAL QUANTITATIVE INVESTING CHALLENGE 2026

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Strategy Declaration

ML Supertrend: Adaptive Reversal Strategy

Young Investors Cup 2026 gold medallists

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Investment hypothesis and complete strategy

The strategy seeks short-term rebounds after unusually strong selling in volatile stocks. A fresh low alone is not a BUY: price must confirm an upward Supertrend reversal, recent oversold RSI and signal spacing. The machine-learning layer learns online from completed hypothetical long and short probes. Three model components use rolling outcomes, adverse-movement-weighted batches and an 8 by 8 market-context memory to adjust the Supertrend multiplier and ATR smoothing within bounds. The learning changes signal sensitivity and timing; explicit portfolio rules still set entry, position size and exit. The hypothesis fails when selling reflects lasting information, or costs erase any rebound.

Each probe opens at a candle open and matures at the close of its fifth candle. A move of one entry ATR gives the long probe a score of +1 and the matching short probe -1. Only matured outcomes can update later settings. Because probes overlap, their labels are dependent and are not account returns.

At equal confidence, one ATR of adverse movement halves a probe's batch weight. Context-memory influence rises with its supporting observations but is capped at 65%. A bounded update layer smooths the models' parameter proposals; it does not forecast a stock's return or decide how much capital to allocate. Position size follows separate risk limits.

Why these stocks

DELL is the paper-trading example. For risk calibration, we use a separate historical cross-sector sample. It is not the strategy portfolio or a list of declared holdings. The trading universe remains broader: each stock must pass its BUY, liquidity and risk checks before receiving capital.

1 Strategy and quantitative method

Use completed regular-session 30-minute candles for supported S&P 500 stocks. BUY opens one long position; repeated BUYs do not add size. SELL or an earlier protective stop closes it. Orders follow confirmation; historical replay fills at the next bar open plus costs.

1. Measure normal price movement

True Range Equation

$$TR_t = \max\{H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|\}$$

t is the current completed candle; t - 1 is the previous candle. H, L and C are high, low and close in dollars. TR takes the largest candle range or gap from the previous close. Vertical bars mean absolute value.

Volatility Envelope Equations

$$x_t = \frac{H_t + L_t + 2C_t}{4}, \qquad A_t = A_{t-1} + \frac{TR_t - A_{t-1}}{\ell_t}$$

$$b_t^- = x_t - m_t A_t, \qquad b_t^+ = x_t + m_t A_t$$

x is the close-weighted price, A the smoothed true range, l the smoothing length and m the multiplier. The boundaries sit mA above and below x; the baseline uses l = 24 and m = 1.4. They trail or reset around price. A close across the previous opposing band flips trend.

2. Machine-learning models and completed probes

Probe Return Equation

$$g = s\frac{C - P}{A_{\text{entry}}}$$

g is the probe score; s is +1 for a long and -1 for a short. P is entry price, C the fifth-candle close and A entry the smoothed true range at entry. A one-ATR price rise gives g = +1 for the long and -1 for the short. Up to five probes per direction can overlap; each result enters learning after its fifth candle closes.

Three online model components use matured probes. The rolling-performance learner tracks a 30-outcome baseline window per direction: below 38% wins with negative mean score, it proposes wider bands and slower ATR smoothing; above 62% with a positive mean, tighter and faster settings. The micro-batch learner weighs each result by confidence divided by one plus adverse excursion in ATR units, then assesses normalized gain, profit factor and downside ratio. The context-memory learner maps trend/range state and relative volatility to an 8 by 8 grid. Hurst persistence, inverse entropy and ADX contribute to the regime score; smoothed cell proposals blend with Gaussian-distance-weighted neighbors. Its influence grows with sample count, capped at 0.65.

A meta-combiner blends rolling and context proposals, then incorporates available batch recommendations. A larger multiplier requires a larger reversal to flip trend; a longer ATR length reacts more slowly to volatility. The smoothed update changes signal timing, not account exposure. Deadbands, per-update caps, bounded parameter ranges and periodic baseline reversion limit drift. These are adaptive online-learning rules, not a neural network or evidence of predictive edge.

Adaptive Update Equation

$$\theta_{t+1} = \theta_t + \alpha_t(\tilde{\theta}_t - \theta_t)$$

theta is the current multiplier or ATR length; theta tilde is the rounded, bounded target; alpha is the update fraction. For multiplier 1.40, target 1.50 and alpha 0.20, the update is 1.42. Deadbands ignore tiny changes; caps constrain each update and periodic reversion pulls settings toward baseline. The multiplier stays 0.5-5.0 and ATR length 5-100.

3. Turn the adjusted indicator into a decision

BUY requires a stored fresh low, an upward trend flip, RSI(14) below 30 within 50 candles and at least 10 candles since the last signal. RSI compares smoothed gains and losses on a 0-100 scale. SELL uses a fresh-high reversal, downward flip and recent RSI above 70. Volume-surge filtering and live tick pressure are off; all decisions use confirmed candles.

Learning check. Across 780 DELL bars, 776 probes per side matured. Multiplier moved 1.40-2.145 while ATR stayed at 24; adaptive signals were 6 BUY / 8 SELL versus 7 / 10 fixed. Prefix checks matched. This verifies implementation, not return edge. Data: Yahoo Finance.

2 Adaptive model test and backtest

To test the contribution of online learning, we replayed the declared 30-minute engine on 17 selected stocks plus SPY from 2 July to 9 October 2026 (70 sessions). Both strategy portfolios use identical entry, exit and risk rules; the fixed baseline disables learning and holds ATR(24) and multiplier 1.4. Signals are processed sequentially after completed bars and filled at the next available open, with adverse friction of 5 or 10 basis points per side.

Portfolio	Friction	Return	Max DD	Sharpe (0% RF)	Trades
Adaptive ML settings	5 bp	+5.95%	-3.71%	1.94	70
Fixed ATR(24) / 1.4	5 bp	+12.97%	-3.89%	2.97	97
75% SPY / 25% cash	5 bp	+3.08%	-2.52%	1.39	-
Adaptive ML settings	10 bp	+4.22%	-3.80%	1.40	72
Fixed ATR(24) / 1.4	10 bp	+11.50%	-3.77%	2.63	94
75% SPY / 25% cash	10 bp	+3.04%	-2.52%	1.37	-

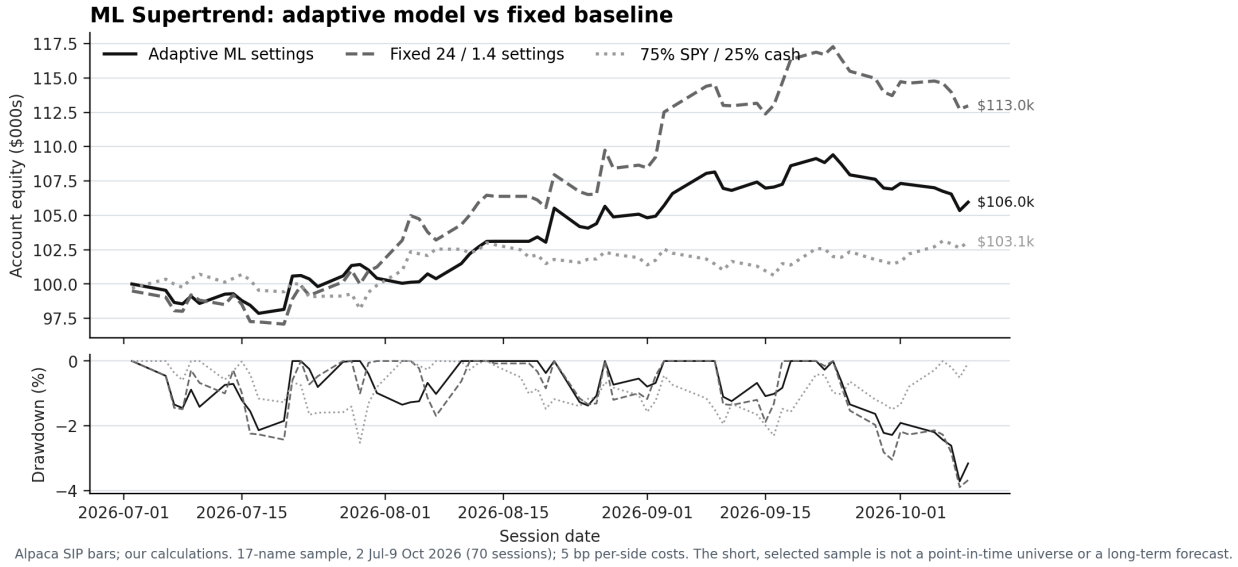


Figure 1. Sequential 30-minute replay: adaptive model, fixed-parameter baseline and cash-matched benchmark. Data: Alpaca SIP bars; our calculations.

The adaptive settings beat the benchmark in this sample but lagged the fixed baseline at both cost levels. This test therefore does not show that the learning layer improved returns. Average adaptive exposure was 22.06%, below the 75% ceiling because stop-risk and concentration limits often reduced position size. Its 6% drawdown halt did not trigger. The selected 17-stock sample and 70-session window do not establish a durable edge or long-term performance.

Replay code, trade ledger, source hashes and monthly results: direct model backtest.

3 Trade expectancy and risk tests

Net Trade Expectancy Equation

$$e = pW - (1-p)D, \qquad p_{\mathrm{BE}} = \frac{D}{W+D}$$

e is mean net return per closed trade; p is the net win rate; W is the average net winning return and D the positive magnitude of the average net losing return. All returns use percentage units and already include costs. The break-even win rate, p BE, is the win rate needed to offset losses at those payoff sizes.

Trade statistic	Historical result
Completed trades	622
Win rate	46.62%
Mean net profit / return per trade	+$81.73 / +0.8173%
Median trade return	-0.6268%
Average net winning / losing return	+8.05% / -5.50%
Profit factor	1.28
Break-even win rate at observed payoffs	40.59%
Worst trade	-40.03%
Longest losing streak	15 trades

These trade statistics come from the separate archived email-ranking replay, not the declared 30-minute model backtest. Within that earlier sample, the win rate exceeded the 40.59% break-even level implied by observed payoff sizes; profit factor was 1.28. The median trade still lost 0.6268%, showing that the mean gain was lifted by a small number of larger wins.

The 15-loss streak and -40.03% worst trade show why position limits and gap controls matter. These are 622 closed trades from the earlier email replay; ten open positions are excluded. Streaks follow recorded SELL order, including same-day exits. Calculations: GitHub trade evidence.

Monte Carlo: exposure, correlation and gaps

The separate historical risk sample is distinct from the five-position strategy portfolio. We fit its covariance to 59 aligned daily log returns, 2 July-25 September 2026, with 11.00% Ledoit-Wolf shrinkage. Each model runs 50,000 fifteen-session paths at 40%, 60% and 75% exposure, with fixed quantities and zero cash interest. These scenarios compare risk levels; they are not strategy returns. Expected shortfall (ES) is the average account loss in the worst 5% of paths.

Risk scenario	ES 95% at 40% / 75%	Investment implication
Gaussian / Student-t(5)	3.50% / 6.56%; 3.56% / 6.68%	Zero-drift baseline / heavier tails.
Five-session block bootstrap	2.92% / 5.48%	Preserves sampled return blocks; unseen crashes remain possible.
Volatility x 1.5; correlation 0.90	9.36% / 17.55%	Correlated losses are the dominant portfolio stress.
Common 20% opening gap	10.81% / 20.27%	At 75%, daily-close liquidation lowers ES to 16.44%.

Data: Yahoo Finance daily closes; our calculations. Common dates only, no missing-price fills. Gaussian ES rises from 3.50% at 40% to 6.56% at 75%; the 60% comparison is 5.25%. At 75%, 7.18% of Gaussian paths reach the 6% drawdown trigger. These fixed-holding scenarios test risk, not strategy returns. Gaussian expected price drift is zero; the short calibration leaves regime uncertainty. Seeds: 20261102-20261106.

4 Portfolio construction and risk

False reversals can cluster during persistent downtrends; overnight gaps can cross stops, and technology names can fall together. Entry spacing, planned-loss budgets, sector/theme caps and the account halt address these risks. All limits below apply together and are declared design choices, not optimized thresholds.

Control	Execution rule and purpose
Exposure	Long-only, no borrowing; at most five names, 15% of equity per name and 75% gross stocks. Five names at the individual cap use the full portfolio allowance.
Loss budget	Initial stop two 30-minute ATR(24) below fill; planned risk at most 0.5% of equity per name and 2% across the account. Never widen a stop.
Concentration	At most 20% in one sector and 25% in the combined semiconductor / technology-infrastructure cluster. Correlated names consume shared capacity.
Liquidity	Price at least $5; trailing 20-session average dollar volume at least $20 million. Reject stale, halted or incomplete quotes.
Account losses	At 2% session loss, pause entries until the next session. At 6% from peak marked equity, liquidate at available prices and halt for the competition.
Execution integrity	Use one account; reject duplicate BUY entries. Log accepted/rejected signals, settings, quotes, order and fill times, fees, stop and exit reason. Freeze the declared rules; close by the competition cutoff.

Position Sizing Equation

$$d_i = 2\,\mathrm{ATR}_{24,i}, \qquad q_i = \left\lfloor \min\left(\frac{0.15E}{P_i}, \frac{0.005E}{d_i}\right) \right\rfloor$$

For stock i, E is marked account equity, P its entry price, ATR with subscript 24 its 24-candle smoothed true range, d the dollar stop distance and q whole shares. Floor brackets round down. The smaller value/risk cap governs; remaining exposure, sector capacity, aggregate risk and costs can reduce the order.

At the recorded DELL close of $563.17, a $15,000 cap permits 26 shares ($14,642.42). The $500 loss budget in a $100,000 account allows at most $19.23 stop distance at that size. A wider two-ATR stop reduces shares. The 15% name limit and five-position limit allow at most 75% gross exposure; remaining equity stays in cash. Position size can be lower when the stop-risk budget binds.

Gap and execution stress

Stop execution stress	Loss / planned stop loss	Account loss at 2% risk budget
Fills at intended stop	1.0 x	2%
Loss exceeds stop distance by 50%	1.5 x	3%
Loss doubles the intended distance	2.0 x	4%
Gap produces three times planned loss	3.0 x	6%

Calculation: aggregate planned loss multiplied by the realized loss/stop-distance ratio. This is a deterministic exposure test, not a forecast. Gaps can exceed all ratios shown; neither a stop nor the 6% liquidation trigger guarantees a maximum loss.

Stress grid: 96 combinations of horizon, exposure, volatility and correlation (100,000 paths each); daily Gaussian, t(5), gap and liquidation tests use 40,000 paths per model. At the 75% ceiling, common 10%/20% falls imply 7.5%/15% account losses before costs. A 6% halt is a trigger, not a loss guarantee.

Final declared strategy

Trade confirmed 30-minute ML Supertrend reversals, long-only. Matured probes update bounded settings; qualifying BUYs pass the sizing and portfolio limits above. Hold at most five names, 15% per name and 75% total stocks. Exit on SELL or stop; pause entries at 2% session loss and liquidate and halt at 6% account drawdown.